Oluwatosin Ayeni

Senior Machine Learning Research Engineer — Microsoft Research Cambridge

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PROFILE

AI & ML Engineer specialising in LLM training, fine-tuning, inference, and evaluation at GPU scale. Full pipeline experience: LoRA/PEFT on Phi-4, Qwen3, Mistral 7B — data prep, training loop, evaluation, production deployment. H100 NVL benchmarks: 5,965 tok/s peak, 1,443× TTFT speedup via RadixAttention, 100% SLO compliance. Built automated benchmark harness spanning closed APIs and open serving engines. MSc Artificial Intelligence, Distinction, University of Strathclyde 2023. Open to relocating to Cambridge.

CORE COMPETENCIES

ML Training & Fine-tuning	PyTorch, PEFT/LoRA, Mistral 7B, Phi-4, Qwen3, Gemma 2 — full pipeline: data prep, training loop, eval, serving; multi-GPU (TP=2/4)
Inference & Serving	vLLM, SGLang, TensorRT-LLM, KV cache optimisation, speculative decoding, continuous batching, prefix caching, AWQ/GPTQ
Evaluation & Benchmarking	Automated harness across closed/open APIs; TTFT, TPOT, throughput, SLO metrics; GPU profiling with NVIDIA Nsight
GPU & Systems	CUDA kernel profiling, memory bandwidth analysis, warp occupancy, BFloat16/FP8, multi-GPU parallelism, RTX 4090 and H100 NVL
Infrastructure	Kubernetes (HPA, GPU pools), Docker, Prometheus, Grafana, FastAPI, Python, GitHub Actions CI/CD

PROFESSIONAL EXPERIENCE

Jan 2025–present — **Founder & Lead Engineer** — *Nexora Academic*, Glasgow, UK

- H100 NVL benchmark: vLLM v0.30 vs SGLang v0.5.20 — Qwen3-8B BF16, TP=2. Peak: 5,965 tok/s; RadixAttention: 1,443× TTFT speedup (39 ms vs 55,565 ms) at 131K-token context; 100% SLO compliance up to concurrency 128. Live: inference-web-theta.vercel.app
- Built *inference-optimizer*: production AI Inference Gateway — 3.87× faster (parallel execution), 61.4% cost reduction (prompt caching), 14× throughput (concurrency), 64.3% cheaper (model routing). 13 optimisations, 4 providers.
- Built *vllm-k8s-stack*: Kubernetes LLM serving — KV-cache-aware routing, HPA autoscaling on GPU pools, Prometheus/Grafana observability.
- Built *inference-harness*: unified benchmark framework spanning closed APIs + vLLM/SGLang/TensorRT-LLM with GPU profiling and live dashboard.
- Designed 13-phase AI Inference Engineering curriculum covering GPU architecture, CUDA optimisation, serving frameworks, and distributed inference.

Jan 2023–present — **Founder & AI Consultant** — *Vertinab / AnaburyAI Technology*, Glasgow, UK

- 28× LLM cost reduction deploying Qwen2.5-7B on RTX 4090 via vLLM/SGLang with continuous batching and KV cache optimisation.
- CUDA KV cache kernel profiling with NVIDIA Nsight: identified memory bandwidth bottleneck; 3.8× throughput improvement; 18× latency reduction via continuous batching redesign.
- Fine-tuned Mistral 7B with LoRA/PEFT on radiology corpus: +22% diagnostic accuracy, −35% hallucination rate; deployed as production clinical inference API.

2024–present — **Senior Software Engineer (Contract)** — *EduTechConnect / Stellar Labs*, Remote

- Real-time NLP pipeline (WhatsApp → Zoho CRM → Supabase) with sub-500 ms end-to-end latency under concurrent load.

Mar 2026–present **AI Automation Engineer** — *EduTech Global*, Remote

- LLM-backed student chatbots with structured output extraction and multi-step reasoning for Ahmadu Bello and Babcock universities.

2018–2022 **Data Analyst** — *Bytelab Technologies Ltd*, Nigeria

KEY PROJECTS

llm-inference-benchmark (H100) vLLM v0.30 vs SGLang v0.5.20 on 2× H100 NVL 94 GB, Qwen3-8B BF16, TP=2. RadixAttention: 1,443× TTFT speedup. Peak 5,965 tok/s. SLO 100% at concurrency 128. Live results

My_finetuned_Phi4 Phi-4 LoRA fine-tuning on Cosmopedia: full pipeline — data prep, LoRA adapter training, evaluation metrics, inference benchmarking. GitHub

Qwen3-TinyStories Complete Qwen3-based LM: tokeniser training, architecture config, training loop, generation evaluation. GitHub

Offline AI Health Assist. Mistral 7B fine-tuned on radiology corpus: +22% diagnostic accuracy, −35% hallucination rate; production clinical inference API. GitHub

inference-harness Unified benchmark harness: closed + open serving engines (vLLM, SGLang, TensorRT-LLM), GPU profiling (Nsight), automated evaluation dashboard. GitHub

EDUCATION

2022–2023 **MSc Artificial Intelligence and Applications, Distinction** — *University of Strathclyde*, Glasgow, UK

2010–2016 **MTech / BSc Physics** — *Federal University of Technology Akure*, Nigeria

OPEN SOURCE

GitHub 43 public repos — github.com/AyeniOluwatosinOlawale